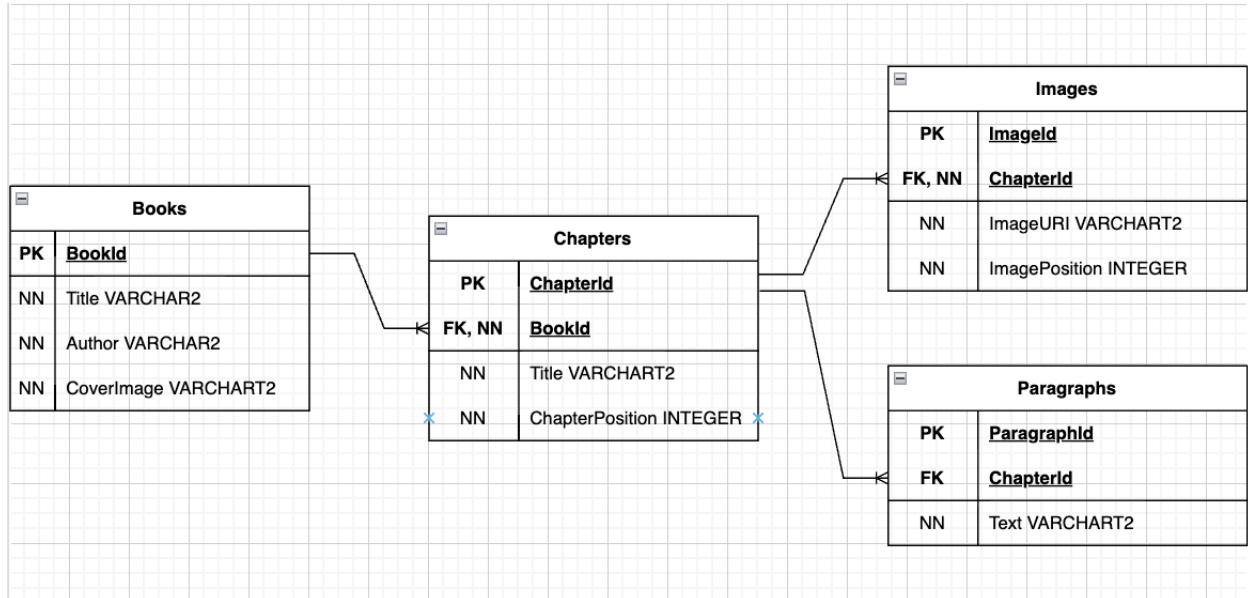


Data Model:



Entities & Relationships

1. Books Table

This table stores general information about each book, including the title, author, and URI for the cover image.

a. **Table Name:** Books

b. **Columns:**

- book_id (INTEGER, Primary Key, Auto-Increment) – Unique identifier for each book.
- title (TEXT) – Title of the book.
- author (TEXT) – Author of the book.
- cover_uri (TEXT) – URI to the cover image of the book.

2. Chapters Table

This table stores chapter-related information, including the title and the chapter's position in the book.

a. **Table Name:** Chapters

b. **Columns:**

- chapter_id (INTEGER, Primary Key, Auto-Increment) – Unique identifier for each chapter.
- book_id (INTEGER, Foreign Key, References Books(book_id)) – Links each chapter to a specific book.

- iii. title (TEXT) – Title of the chapter.
- iv. position (INTEGER) – Position of the chapter in the book (e.g., 1 for the first chapter, 2 for the second, etc.).

3. Images Table

This table stores the images associated with each chapter, including their positions within the chapter.

a. **Table Name:** Images

b. **Columns:**

- i. image_id (INTEGER, Primary Key, Auto-Increment) – Unique identifier for each image.
- ii. chapter_id (INTEGER, Foreign Key, References Chapters(chapter_id)) – Links each image to a specific chapter.
- iii. image_uri (TEXT) – URI to the image file.
- iv. position (INTEGER) – Position of the image within the chapter (e.g., image 1, image 2, etc.).

4. Paragraphs Table

This table stores the text of the paragraphs for each chapter.

a. **Table Name:** Paragraphs

b. **Columns:**

- i. paragraph_id (INTEGER, Primary Key, Auto-Increment) – Unique identifier for each paragraph.
- ii. chapter_id (INTEGER, Foreign Key, References Chapters(chapter_id)) – Links each paragraph to a specific chapter.
- iii. text (TEXT) – The actual content of the paragraph.

SQL Queries

1. Insert entities into the Room Database using @Insert

Example:

@Insert

fun insertBook(book: Books)

2. Get all books

This query retrieves all books, including the title, author, and cover image URI.

SELECT book_id, title, author, cover_uri FROM Books;

3. Find a book for a given ID

This query retrieves all information of a book with a specific id

SELECT * FROM books WHERE bookId=:id

4. Delete a book with a given id

This query will be used to delete a book with a specific id

DELETE FROM books WHERE bookId=:id

5. Get chapters of a specific book

This query retrieves all chapters for a specific book, ordered by their position.

SELECT chapter_id, title, position FROM Chapters WHERE book_id = :book_id ORDER BY position;

6. Find a chapter for a given ID

This query retrieves all information of a chapter with a specific id

SELECT * FROM chapters WHERE chapterId = :id

7. Delete a chapter with a given id

This query will be used to delete a chapter with a specific id

DELETE FROM chapters WHERE chapterId = :id

8. Get paragraphs of a specific chapter

This query retrieves all paragraphs for a specific chapter.

SELECT paragraph_id, text FROM Paragraphs WHERE chapter_id = :chapter_id ORDER BY paragraph_id;

9. Find a paragraph for a given id

This query retrieves all information of a paragraph with a specific id

SELECT * FROM paragraphs WHERE paragraphId = :id

10. Delete a paragraph with a given id

This query will be used to delete a paragraph with a specific id

DELETE FROM paragraphs WHERE paragraphId = :id

11. Get images of a specific chapter

This query retrieves all images for a specific chapter, ordered by their position.

**SELECT image_id, image_uri, position FROM Images WHERE chapter_id = :chapter_id
ORDER BY position;**

12. Find an image for a specific id

This query retrieves all information of an image with a specific id

SELECT * FROM images WHERE imageId = :id

13. Delete an image with a given id

This query will be used to delete an image with a specific id

DELETE FROM images WHERE imageId = :id

Testing Strategy

Instrumentation Testing

Navigation Testing:

For navigation components like the Bottom Bar, Navigation Rail, and Navigation Drawer, we will write tests to simulate user actions and check if the expected screen is displayed.

Example:

Test Case 1: When the user clicks the "Library" button in the Bottom Bar, ensure that the Library screen is displayed.

UI Element Testing:

We will also test that UI elements like buttons, text fields, and images render correctly. This includes verifying the layout of paragraphs and images within each chapter.

Example:

Test Case 2: Verify that the chapter's content (text and images) is displayed correctly, with proper formatting and order.

Unit Testing

ViewModel Testing:

We will write tests for the ViewModels to ensure they correctly manage and provide data for UI components.

Example:

Test Case 1: Test that the ViewModel correctly loads and provides a list of books.

Test Case 2: Test that the ViewModel correctly loads chapters and paragraphs for a given book.

Data Layer Testing:

We will test the Room database operations to ensure data is being saved, updated, and retrieved correctly.

Example:

Test Case 3: Test that a book can be inserted into the database and retrieved with the correct attributes (title, author, cover image).

Business Logic Testing:

Ensure that state changes are correctly reflected in the app, especially when modifying the content of books or chapters.

Example:

Test Case 4: Test that adding a new chapter to a book correctly updates the UI and the database.